25/10/2019

UNIT - IV

Syllabus: Dimensionality Reduction - Linear Discriminant Analysis, Principal Component Analysis, factor Analysis, Independent Compont Analysis, Locally Linear Embedding, Isomap, least Square optimization

Evolutionary learning, Genetic algorithms, Genetic offspring, Genetic operators, using Genetic algorithms.

Dimensionality Reduction:-

In many learning problems, the datasets have large number of variables.

sometimes, the no of variables is more than the no of observations.

for example, in many Scientific field such as

- Image processing
- time Series analysis
- Internet Search engines &c
- Automatic text analysis

- Statistical and ML methods have some difficulty when dealing with such high dimensional data
- Normally the no of variables (input) is reduced before the ML alg can be Succesfully applied.

The no of input features, variables or columns Present in a given dataset is known as dimensionality, and the proces to reduce these features is called dimensionality reduction.

Dimensionality reduction technique can be defined as 'It is a way of converting the higher dimensions dataset into lesser dimensions dataset ensuring that it provides similar info

Dimensionality Reduction is implemented in two ways

① feature selection

we are interested in finding 'k' of the total 'n' features that gives us the most info and we discard the other (n-k) dimensions.

Process of selecting the subset of the relevant features and leaving out the irrelevant features present in a dataset to build a model of high accuracy.

② feature Extraction

⇒ we are interested in finding a new set of 'k' features that are combination of the original 'n' features

It is the process of transforming the space containing many dimensions into space with fewer dimensions. This approach is useful when we want to keep the whole info but fewer resources while performing processing the info.

These methods may be supervised or unsupervised depending on whether or not they use the o/p info.

The best known and most widely used feature extraction methods are PCA & LDA.

we need to check whether the particular feature producing any error or not. Based on how much error it is producing.

In both methods we require a measure of the error in the model.

In regression problems, we may use the

- Mean Squared Error (MSE) or
- Root mean Squared Error (RMSE)

It is the sum, over all the dp, of the square of the distance b/w the predicted and actual target variables, divided by the no of datapoints

If $y_1, y_2 \dots y_n$ are observed values and $\hat{y}_1, \hat{y}_2 \dots \hat{y}_n$ are the predicted values then

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

n - Avg of all the Examples

$i=1$ - no of datapoints

y_i - Actual target

$\hat{y}_i$ - predicted target.

In classification problems, we may use the misclassification rate as a measure of the error.

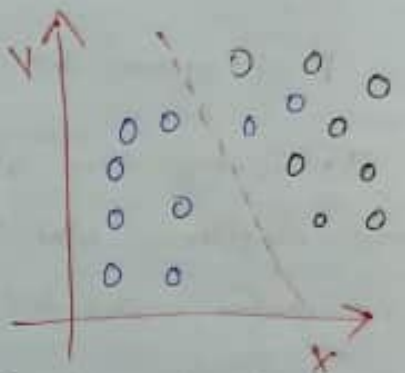
This is defined as follows:

$$\text{Misclassification rate} = \frac{\text{no. of misclassified examples}}{\text{total no of examples}}$$

Linear Discriminant Analysis (LDA):-

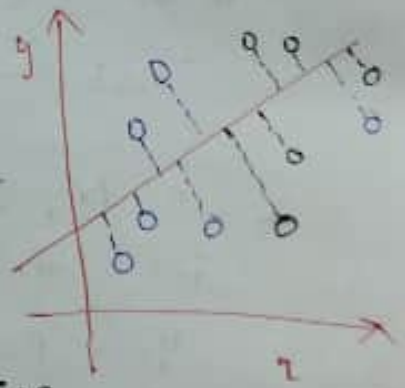
It is also called as Discriminant functional Analysis, or Normal Discriminant Analysis is a dimensionality Reduction technique that is Commonly used for Supervised classification problems.

It is used to project the features in higher dim Space into a lower dimension space.



Suppose, we have two sets of dp belonging to two diff classes that we want to classify.

- when the dp are plotted on the 2D plane, there's no straight line that can separate the two classes of the datapoints completely.



- Here, LDA uses both axes (x & y) to create a new axis and projects data onto a new axis in a way to maximize the separation of the two categories and hence, reducing the 2D graph into a 1D graph.

But, whenever we want to create 2D data to 1D data we need to Consider two criteria by LDA to create a new axis.

① Maximize the distance b/w means of two classes.

② Minimize the variation within each class.



Now discuss, the steps to convert Higher data to lowest dim data.

① Compute the class means of dependent variable

$$\mu_1 = \frac{1}{N_1} \sum_{x \in \omega_1} x \quad \omega_1 \rightarrow \text{class}$$

② Derive the Covariance matrix of the class variable
(measure of r/s b/w two random var and how they change together.)

$$S_1 = \sum_{x \in \omega_1} (x - \mu_1)(x - \mu_1)^T$$

all examples present in the class

③ Compute the within class scatter matrix (sum of all Covariance matrix)

$$S_W = S_1 + S_2$$

④ Compute the b/w class scatter matrix

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

⑤ Compute Eigen values and eigen vectors from the within class and b/w class scatter matrix.

$$S_W^{-1} S_B w = \lambda w$$

⑥ Sort the values of eigen values and select the top k values [sort in decreasing order]

⑦ find the eigen vectors corresponds to the top k eigen vectors

$$\left[S_W^{-1} S_B - \lambda I \right] \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = 0$$

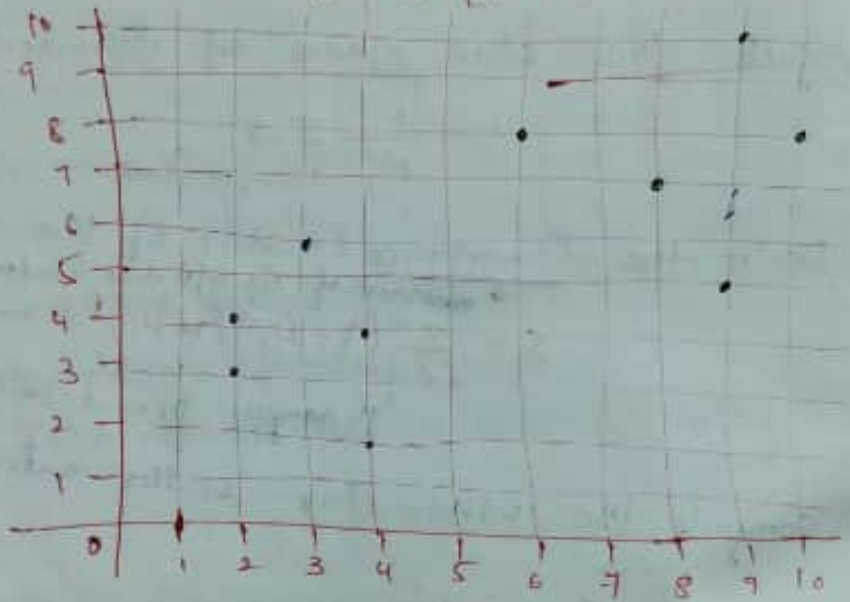
w_1, w_2 [notational representation]

⑧ Obtain the LDA by taking the dot product of eigen vectors and original data.

eg: Compute the LD projection for the following two dimensional dataset

Samples for class ω_1: $X_1 = (x_1, x_2) = \{(4,2), (2,4), (2,3), (3,6), (4,4)\}$

Samples for class ω_2: $X_2 = (x_1, x_2) = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$



① The class mean are:

$$\mu_1 = \frac{1}{N_1} \sum_{x \in \omega_1} x = \frac{1}{5}\left[\begin{pmatrix}4\\2\end{pmatrix} + \begin{pmatrix}2\\4\end{pmatrix} + \begin{pmatrix}2\\3\end{pmatrix} + \begin{pmatrix}3\\6\end{pmatrix} \begin{pmatrix}4\\4\end{pmatrix}\right] = \begin{bmatrix}3\\3.8\end{bmatrix}$$

$$\mu_2 = \frac{1}{N_2} \sum_{x \in \omega_2} x = \frac{1}{5}\left[\begin{pmatrix}9\\10\end{pmatrix} + \begin{pmatrix}6\\8\end{pmatrix}\begin{pmatrix}9\\5\end{pmatrix} + \begin{pmatrix}8\\7\end{pmatrix} + \begin{pmatrix}10\\8\end{pmatrix}\right] = \begin{bmatrix}8.4\\7.6\end{bmatrix}$$

② Covariance matrix of the first class:

$$S_1 = \sum_{x \in \omega_1} \frac{(x-\mu_1)(x-\mu_1)^T}{N-1}$$

$$= \left[\begin{pmatrix}4\\2\end{pmatrix} - \begin{pmatrix}3\\3.8\end{pmatrix}\right]\left[\begin{pmatrix}4\\2\end{pmatrix} - \begin{pmatrix}3\\3.8\end{pmatrix}\right]^T + \left[\begin{pmatrix}2\\4\end{pmatrix} - \begin{pmatrix}3\\3.8\end{pmatrix}\right]\left[\begin{pmatrix}2\\4\end{pmatrix} - \begin{pmatrix}3\\3.8\end{pmatrix}\right]^T +$$

$$\left[\begin{pmatrix}2\\3\end{pmatrix}-\begin{pmatrix}3\\3.8\end{pmatrix}\right]\left[\begin{pmatrix}2\\3\end{pmatrix}-\begin{pmatrix}3\\3.8\end{pmatrix}\right]^T+\left[\begin{pmatrix}3\\6\end{pmatrix}-\begin{pmatrix}3\\3.8\end{pmatrix}\right]\left[\begin{pmatrix}3\\6\end{pmatrix}-\begin{pmatrix}3\\3.8\end{pmatrix}\right]^T+$$

$$\left[\begin{pmatrix}4\\4\end{pmatrix}-\begin{pmatrix}3\\3.8\end{pmatrix}\right]\left[\begin{pmatrix}4\\4\end{pmatrix}-\begin{pmatrix}3\\3.8\end{pmatrix}\right]^T \Big/ N-1$$

$$= \begin{bmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{bmatrix}$$

$$\begin{bmatrix}1\\-1.8\end{bmatrix}\begin{bmatrix}1 & -1.8\end{bmatrix}$$

$$\begin{bmatrix}1 & -1.8\\ -1.8 & 3.24\end{bmatrix}$$

$$S_2 = \sum_{x \in w_2} \frac{(x-\mu_2)(x-\mu_2)^T}{N-2}$$

$$= \begin{bmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{bmatrix}$$

③ within class scatter matrix:

$$S_w = S_1 + S_2$$

$$= \begin{bmatrix} 1 & -0.25 \\ -025 & 2.2 \end{bmatrix} + \begin{bmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{bmatrix}$$

$$= \begin{bmatrix} 3.3 & 0.3 \\ -0.3 & 5.5 \end{bmatrix}$$

④ $S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$

$$= \left[\begin{pmatrix}3\\3.8\end{pmatrix}-\begin{pmatrix}8.4\\7.6\end{pmatrix}\right]\left[\begin{pmatrix}3\\3.8\end{pmatrix}-\begin{pmatrix}8.4\\7.6\end{pmatrix}\right]^T$$

$$= \begin{bmatrix} -5.4 \\ -3.8 \end{bmatrix}\begin{bmatrix} -5.4 & -3.8 \end{bmatrix}$$

$$= \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix}$$

(5) Find Eigen values.

$S_W^{-1} S_B w = \lambda w$

$\Rightarrow |S_W^{-1} S_B - \lambda I| = 0$ → Cardinality ~~of~~ [determined] ad-bc

$$\Rightarrow \left| \begin{pmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{pmatrix}^{-1} \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{bmatrix} 0.3046 & 0.0166 \\ 0.0166 & 0.1827 \end{bmatrix} \begin{bmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{bmatrix} 9.2213 - \lambda & 6.489 \\ 4.2339 & 2.9794 - \lambda \end{bmatrix} \right| = 0$$

$\Rightarrow (9.2213 - \lambda)(2.9794 - \lambda) - (6.489)(4.2339) = 0$

$\Rightarrow \lambda^2 - 12.2007\lambda = 0$

$\Rightarrow \lambda(\lambda - 12.2007) = 0$

$\Rightarrow \lambda_1 = 0, \quad \lambda_2 = 12.2007$

~~(6)~~ Eigen vector

$$\left[S_W^{-1} S_B - \lambda I \right] \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = 0$$

$$w_1 = \begin{bmatrix} -0.5755 \\ 0.8178 \end{bmatrix} \qquad w_2 = \begin{bmatrix} 0.9088 \\ 0.4173 \end{bmatrix} = w^* \text{ [eigen vector]}$$

or directly,

$$w^* = S_W^{-1} (M_1 - M_2) = \begin{bmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{bmatrix}^{-1} \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]$$

$$= \begin{bmatrix} 0.3045 & 0.0166 \\ 0.0166 & 0.1827 \end{bmatrix} \begin{bmatrix} -3.4 \\ -3.8 \end{bmatrix}$$

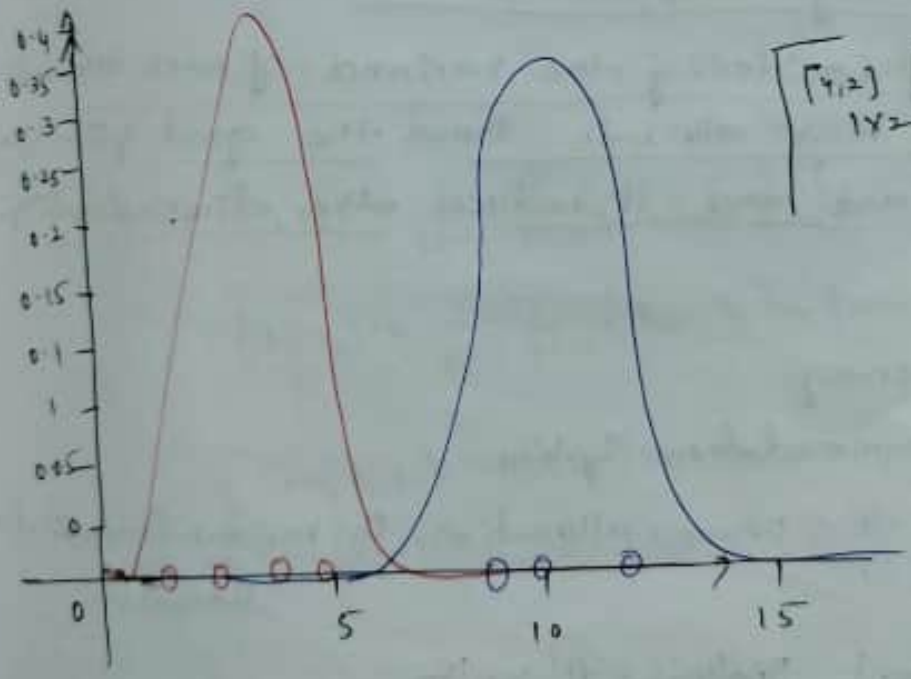
$$= \begin{bmatrix} 0.9088 \\ 0.4113 \end{bmatrix} = w^*$$

(8) obtain the LDA by taking dot product

x_1	4	2	2	3	4	9	6	9	8	10
x_2	2	4	3	6	4	10	8	5	7	8

~~4.46 4.46 3.48 3.06 5.2 5.3 12.35 8~~

y (1) 4.46 3.48 3.06 5.2 5.3 12.35 8 8 10.2 10.19 12.42



$$\begin{bmatrix} [4,2] & \begin{bmatrix} 0.9 \\ 0.4 \end{bmatrix} \\ 1\times 2 & 2\times 1 \\ & 1\times 1 \end{bmatrix}$$

Principal Component Analysis:-

It is an unsupervised learning alg. that is used for the dimensionality reduction in ML. It is a statistical process that converts the observations of correlated features into a set of linearly uncorrelated features with the help of orthogonal transformation.

These new transformed features are called the Principal Components.

It is a technique to draw strong patterns from the given dataset by reducing the variances.

It generally tries to find the lower-dimensional surface to project the high-dimensional data.

It works by considering the variance of each attribute because the high attribute shows the good split b/w the classes, and hence it reduces the dimensionality.

Applications:-

- Image processing
- Movie recommendation system
- Optimizing the power allocation in various communication channels.

Principal Component Analysis - Algorithm

Step 1:- Data

- We consider a dataset having n features or variables denoted by $x_1, x_2 \dots x_n$
- Let there be N examples
- Let the values of the ith feature x_i be $x_{i1}, x_{i2} \dots x_{iN}$

features	Eg1	Ex2	...	ExN
x_1	x_{11}	x_{12}	...	x_{1N}
x_2	x_{21}	x_{22}	...	x_{2N}
⋮	⋮	⋮		
x_i	x_{i1}	x_{i2}	...	x_{iN}
⋮	⋮	⋮		
x_n	x_{n1}	x_{n2}	...	x_{nN}

Step2:- Compute the means of the variables

$$\overline{X_i} = \frac{1}{N}(x_{i1} + x_{i2} + \cdots + x_{iN})$$

Step3: Calculate the Covariance matrix

$$Cov(x_i, x_j) = \frac{1}{N-1}\sum_{k=1}^{N}(x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j)$$

eg:- $$(x_1, x_2) = \frac{1}{N-1}\sum_{k=1}^{N}(x_{11} - \bar{x}_1)(x_{21} - \bar{x}_2)$$

$$(x_{11}, x_2) = \frac{1}{N-1}\sum_{k=2}(x_{12} - \bar{x}_1)(x_{22} - \bar{x}_2)$$

$$S = \begin{bmatrix} Cov(x_1, x_1) & Cov(x_1, x_2) & \cdots & Cov(x_1, x_n) \\ Cov(x_2, x_1) & Cov(x_2, x_2) & \cdots & Cov(x_2, x_n) \\ \vdots & \vdots & & \\ Cov(x_n, x_1) & Cov(x_n, x_2) & \cdots & Cov(x_n, x_n) \end{bmatrix}$$

Step4: Calculate the eigenvalues and eigenvectors of the Covariance matrix.

① Set up the equation:- This is a polynomial eq^n of degree 'n' in. It has n real roots and these roots are the eigenvalues of 'S' $\det(S - \lambda I) = 0$

④ If $\lambda = \lambda'$ is an eigenvalue, then the corresponding eigenvector is a vector

$$U = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{such that} \quad (S - \lambda' I)U = 0$$

⑤ we need to normalize the eigenvectors. Given any vector X we normalize it by dividing X by its length (or, the norm) of the factor.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ is defined as } \|X\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

we compute the n normalized Eigenvectors $e_1, e_2 \cdots e_n$ by

$$e_i = \frac{1}{\|U_i\|} (U_i) \quad i = 1, 2, 3 \cdots n$$

Step5:- Derive new data set

- order the eigenvalues from highest to lowest
- The unit eigenvector Corresponding to the largest eigenvalues is the first principal component.

① let the eigenvalues in descending order be $\lambda_1 \geq \lambda_2 \cdots \lambda_n$ and let the Corresponding unit eigenvectors be $e_1, e_2 \cdots e_n$.

② choose a positive integer p such that $1 \leq p \leq n$

③ choose the eigenvectors Corresponding to the eigenvalues $\lambda_1, \lambda_2, \cdots \lambda_n$ and form the following $p \times n$ matrix

④ form the $n \times N$ matrix

⑤ Next Compute the matrix; $X_{new} = FX$

eg:- Given the data in Table, reduce the dimension from 2 to 1 using the PCA algorithm

feature	Ex1	Ex2	Ex3	Ex4
x_1	x_{11} 4	x_{12} 8	x_{13} 13	x_{14} 7
x_2	x_{21} 11	x_{22} 4	x_{23} 5	x_{24} 14

Step1: Calculate mean

$$\bar{x_1} = \frac{1}{4}(4+8+13+7) = 8$$

$$\bar{x_2} = \frac{1}{4}(11+4+5+14) = 8.5$$

Step2: Cal the covariance matrix

$$S = \begin{bmatrix} Cov(x_1, x_1) & Cov(x_1, x_2) \\ Cov(x_2, x_1) & Cov(x_2, x_2) \end{bmatrix}$$

$$Cov(x_1, x_1) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{1k} - \bar{x_1})(x_{1k} - \bar{x_1})$$

$$= \frac{1}{3}\left((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2\right)$$

$$= 14$$

$$Cov(x_1, x_2) = \frac{1}{3}\left((4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5)\right)$$

$$= \frac{1}{3}(-33) \quad \left[\frac{1}{N-1}\sum_{k=1}^{N}(x_{1k} - \bar{x_1})(x_{2k} - \bar{x_2})\right.$$

$$= -11$$

$$Cov(x_2, x_1) = Cov(x_1, x_2)$$

$$= -11$$

$$Cov(x_2, x_3) = \frac{1}{N-1} \sum_{k=1}^{N} (x_{2k} - \bar{x}_2)(x_{3k} - \bar{x}_3)$$

$$= \frac{1}{3}\left((11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (14-8.5)^2 \right)$$

$$= \frac{69}{3} = 23$$

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Step 3: eigenvalues of the Covariance matrix

The characteristic Eqn of the Cov matrix is

$$0 = \det(S - \lambda I)$$

$$= \begin{vmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{vmatrix}$$

$$ad - bc = (14-\lambda)(23-\lambda) - (-11)(-11)$$

$$= \lambda^2 - 37\lambda + 201$$

$$\Rightarrow D = b^2 - 4ac = (-37)^2 - 4(1)(201)$$

$$= 1369 - 804$$

$$= 565$$

$$\left[\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right.$$
$$\left. a = 1, b = -37, c = 201 \right]$$

$$\lambda = \frac{37 \pm \sqrt{565}}{2} \qquad \sqrt{565} = 23.76$$

$$\lambda_1 = \frac{37 + 23.76}{2} = 30.38$$

$$\lambda_2 = \frac{37 - 23.76}{2} = 6.62$$

Step 4: Computation of the eigenvectors

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix} = (S - \lambda I)U$$

$$= \begin{bmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$= \begin{bmatrix} (14-\lambda)u_1 - 11u_2 \\ -11u_1 + (23-\lambda)u_2 \end{bmatrix}$$

$\Rightarrow (14-\lambda)u_1 - 11u_2 = 0 \Rightarrow (14-\lambda)u_1 = 11u_2$

$\Rightarrow -11u_1 + (23-\lambda)u_2 = 0 \Rightarrow (23-\lambda)u_2 = 11u_1$

} either consider 1

$$\frac{u_1}{11} = \frac{u_2}{14-\lambda} = t$$

$u_1 = 11t$

$u_2 = (14-\lambda)t$

If $t=1$ $\quad u_1 = \begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix} \Rightarrow u_1 = \begin{bmatrix} 11 \\ 14-\lambda_1 \end{bmatrix}$

⇒ Whenever we want to (Consid) calculate the principal components we have to consider the highest eigenvalue

$$\lambda_1 = 30.3849$$

• To find a unit eigenvector, we compute the length of u_1 which is given by,

$$\|u_1\| = \sqrt{11^2 + (14-\lambda_1)^2}$$

$$= \sqrt{11^2 + (14-30.3849)^2}$$

$$= 19.7348$$

$$e_1 = \begin{bmatrix} 11/\|u_1\| \\ (14-\lambda_1)/\|u_1\| \end{bmatrix}$$

$$= \begin{bmatrix} 11/19.7348 \\ (14-30.3849)/19.7348 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

Step 3:- Computation of first principal components

$$e_1^T \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5574 & -0.8303 \end{bmatrix} \begin{bmatrix} x_{11} - \bar{x}_1 \\ x_{21} - \bar{x}_2 \end{bmatrix}$$

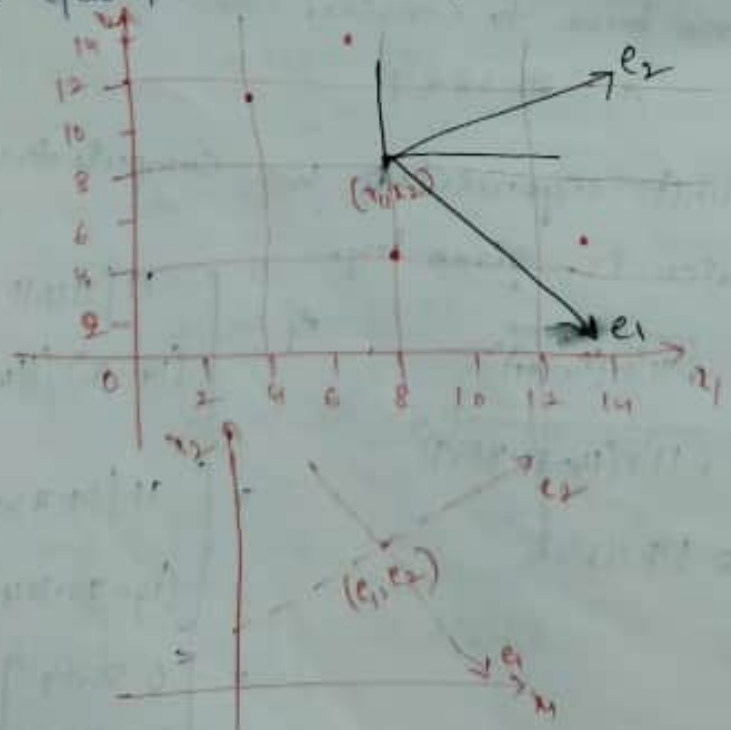
$$= 0.5574(x_{11} - \bar{x}_1) - 0.8303(x_{21} - \bar{x}_2)$$

$$= 0.5574(4 - 8) - 0.8308(11 - 8.5)$$

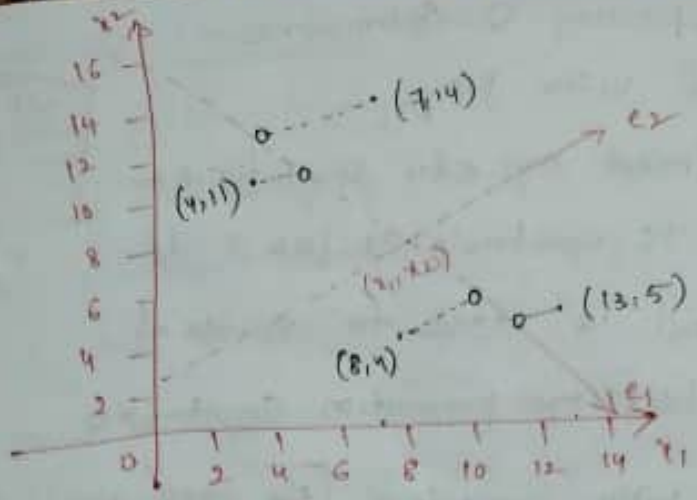
$$= -4.80535$$

feature	Ex_1	Ex_2	Ex_3	Ex_4
x_1	4	8	13	7
x_2	11	4	5	14
1st PC	-4.0352	3.7361	5.6928	-5.1238

Step 4:- Geometrical meaning of first PC



$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5514 \end{bmatrix}$$



Factor Analysis:-

It is a technique that is used to reduce a large no of variables into fewer no of factors.

Factor analysis is a statistical data reduction and analysis technique that strives to explain correlations among multiple outcomes as the result of one or more underlying explanations, or factors.

This technique involves data reduction, as it attempts to represent a set of variables by a small number.

Factor → A Quesstonaeie statement

↓

Two goals - Data reduction and detection of structure.

Data Reduction - reduce large no of variables so that they are economically but accurately represented by a small set of new variables.

Detection of structure - being able to discover an underlying structure. E.g. usefulness is measured by 4 statements.

eg: Job satisfaction Questionnaire

① Satistisfied with pay
- Salary meet my edu Qualification
- salary is equitable to job I do
- Increment is done for appraisal
- Job offers me promotion opportunities

Common factor Analysis

② Satisfied with Supervisor
- Gives me support
- offers me suggestion to improve Quality
- Makes me feel contented over work
- Gives feedback to help me achieve goals.

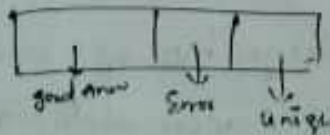
③ Satisfied with Co-workers

④ Satisfied with job

→ Co-variance is due to latent variable that exert Casual influence on observed variables.

Some of the info cannot be explained with a single variable

If any of the single variable is not able explain the Characteristic 100 percent we can called as latent variable.



2factors: Individual Socioeconomic Status and neighborhood Socioeconomic status.

with age sex
- Income
- Educa
- Health fac
- Challenges

10 factors

The whole aim is to reduce the no of factors, to make each factor very clearly with aim & objective

Two types of factor Analysis:-

① Exploratory factor Analysis:-

Measure underlying factors affecting the variables in a data structure without setting any predefined structure to the outcome.

we need to explore when we have a new discovery or invention going on which do not have a priortheory As we opened to as many new factors as possible.

It is a statistical method used to uncover the underlying structure of a relatively large set of variables.

That is when the analyst is not sure how many factors the data consists of ?

It should be used when the researcher has no idea about factors or patterns of measured variables

② Confirmatory factor Analysis:-

- It is usually done when the sample size is very large and try to apply inferential statistics.

- Confirm predefined components already explored implemented for reconfirming the effects and possible correlations Existing b/w a set of predetermined factors and variables.

It is a statistical technique used to verify the factor structure of a set of observed variables.

It allows the researcher to test the hypothesis that a

r/s b/w observed variables and their underlying latent Constructs Exists

[we redo the factor analysis to confirm whether the factors are correct or not.

we have some predefined factors → And there are sets. we can reedit those existing factors.

Two Types

① Traditional Approach

Independent Component Analysis:- [blind source separation technique]

It is a technique used to separate mixed signals into their independent sources.

The application of ICA ranges from audio and image processing to biomedical signal analysis.

What is ICA ?

It is a statistical and computational technique used in ML to separate a multivariate signal into its independent non-Gaussian components.

The goal of ICA is to find a linear transformation of the data such that the transformed data is as close to being statistically independent as possible.

Statistical Independent Concept:-

It is a probability theory that if two random variables X & Y statistically independent.

The joint probability distribution of the pair is equal to

the product of their individual prob distributions, which means that knowing the outcome of one variable does not change the probability of the other outcome.

$$P(X \text{ and } Y) = P(X) * P(Y)$$

(or)

$$P(X \cap Y) = P(X) * P(Y)$$

eg: Simple scenario [cocktail party problem]

① Two people speaking simultaneously in a room.

② Speeches are recorded by two microphones in separate locations.

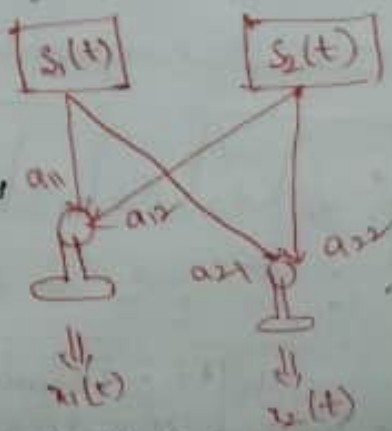
- Let $s_1(t)$, $s_2(t)$ be the speech signals emitted by the two speakers.
- Recorded time signals, by two microphones are $x_1(t), x_2(t)$.
- The recorded time signals can be expressed as a line equation.

Signal emitted with time

Sound recorded by mics

$$x_1(t) = a_{11} s_1(t) + a_{12} s_2(t)$$

$$x_2(t) = a_{21} s_1(t) + a_{22} s_2(t)$$

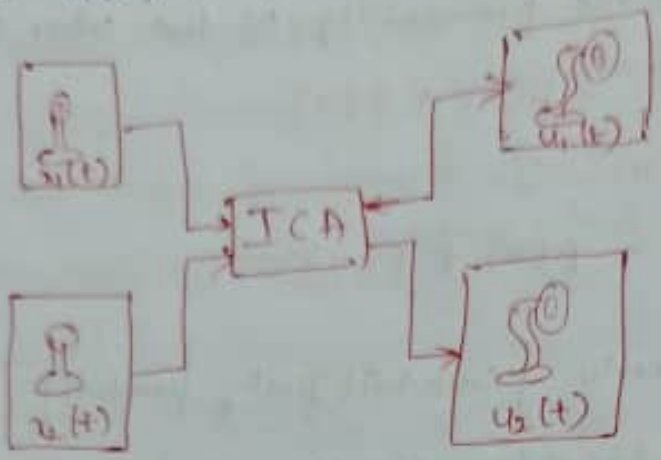


where parameters in matrix A depend on distances of the microphones to the speakers, along with other microphone properties

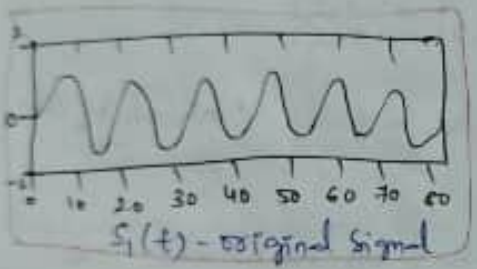
- Assume $s_1(t)$ & $s_2(t)$ are statistically Independent → no r/s b/w s_1 & s_2

Goal: Recover the unmixed speech signals, $s_i(t)$

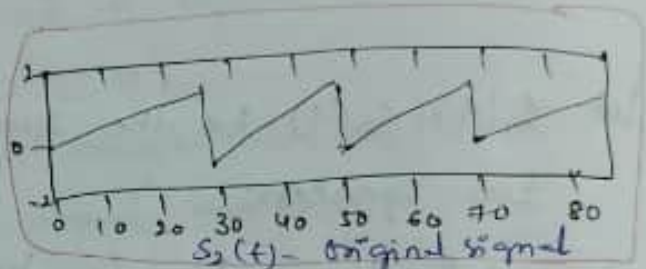
estimate $u_i(t)$, without knowing A or $s_i(t)$



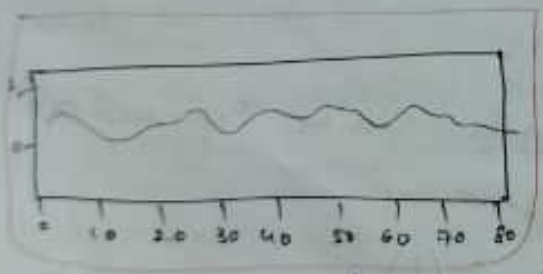
Nature of Signals



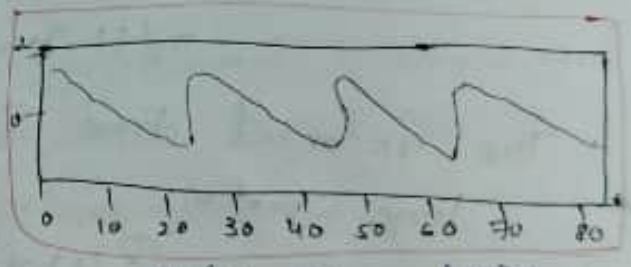
$s_1(t)$ - original signal



$s_2(t)$ - original signal

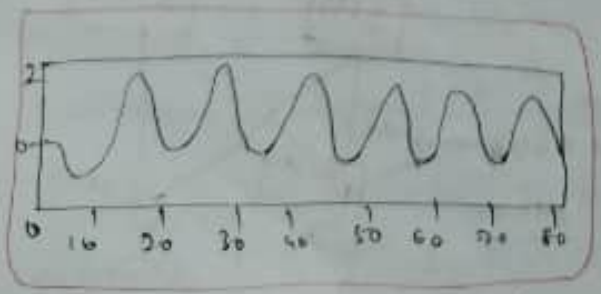


$x_1(t)$ - observed mixture

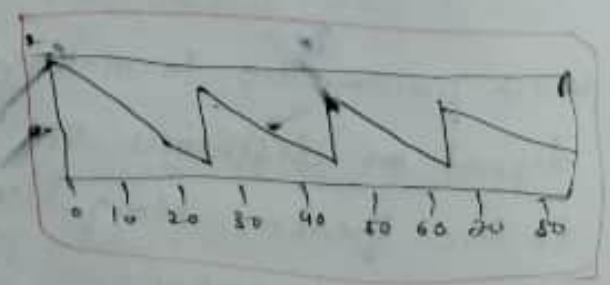


$x_2(t)$: observed mixture.

ICA



$u_1(t)$ - recovered estimate



$u_2(t)$: recovered estimate.

The original signals were very accurately estimated, upto

ICA (Vs) PCA

- Similarity
  - Feature extraction
  - Dimension reduction
- Difference
  - PCA uses up to second order moments of the data to produce uncorelated components.
  - ICA strives to generate components as independent as possible through minimizing both the second order and higher-order dependencies in the given data.

PCA	ICA
- It reduces the dimensions to avoid the problem of overfitting	- It decomposes the mixed signal into its independent sources signals
- It deals with principal components.	- It deals with independent components.
- It focuses on maximizing the variance.	- It doesn't focus on the issue of variance among the data points.
- It doesn't focus on the mutual idependence of the Components	- It focuses on the mutual independence of the Components

Assumptions:

① It asserts that the Source signals are statistically independent to each other.

② The second assumption is that each source signal exhibits non-Gaussian distributions.

Mathematical representation-

- The observed random vector is $X = (x_1, \ldots, x_m)^T$ representing the observed data with m components.
- The hidden components are represented by the random vector $S = (s_1 \ldots s_n)^T$, where 'n' is the hidden sources.

linear static Transformation

The observed data X is transformed into hidden components S using a linear static transformation representation by the matrix W.

$$S = WX$$

W - Transformation matrix

The goal is to transform the observed data X in a way that the resulting hidden components are independent.

Locally linear Embedding:-

It is an unsupervised approach designed to transform data from its original high-dimensional space into a lower-dimensional representation,

→ Firstly, it constructs a nearest neighbors graph to capture these local relationships. Then it optimizes weight values for each data point, aiming to minimize the reconstruction error when expressing a point as a linear combination of its neighbors.

The weight matrix reflects the strength of connections between points.

→ Next, LLE computes a lower dimensional representation of the data by finding eigenvectors of a matrix derived from the weight matrix.

These eigenvectors represent the most relevant directions in the reduced space. Users can specify the desired dimensionality for the output space.

Mathematical Implementation of LLE Algorithm

The key idea of LLE is that locally, in the nearby each data point, the data lies approximately on the linear subspace.

It attempts to unfold the data while preserving these local linear r/s.

Here is the mathematical overview of LLE algorithm.

Minimize: $\Sigma_i \left| x_i - \Sigma_j w_{ij} x_j \right|^2$

Subject to: $\Sigma_j w_{ij} = 1$

where:

x_i - represents the ith data point

w_{ij} - weights that minimize the reconstruction error for data point x_i using its neighbors.



(a) (b) (c)

a) a 2D nonlinear manifold where the data exist on in the 3D original space.

The geodesic distances of points on the manifold are diff from their Euclidian distances.

b) The correct unfolded where the geodesic distances of points on the manifold have been preserved.

c) Applying the linear PCA, which takes Euclidian distances into account, and on the nonlinear data where the subspace has ruined the manifold so the far away red and grey points have fallen next to each other.

Goal of LLE:

LLE unfolds the nonlinear manifold in a piece-wise manner. Each piece is unfolded and the unfolded pieces are put together to have the entire unfolded manifold.



If two points are similar, they should be similar in the subspace. If they are dissimilar, do not care.

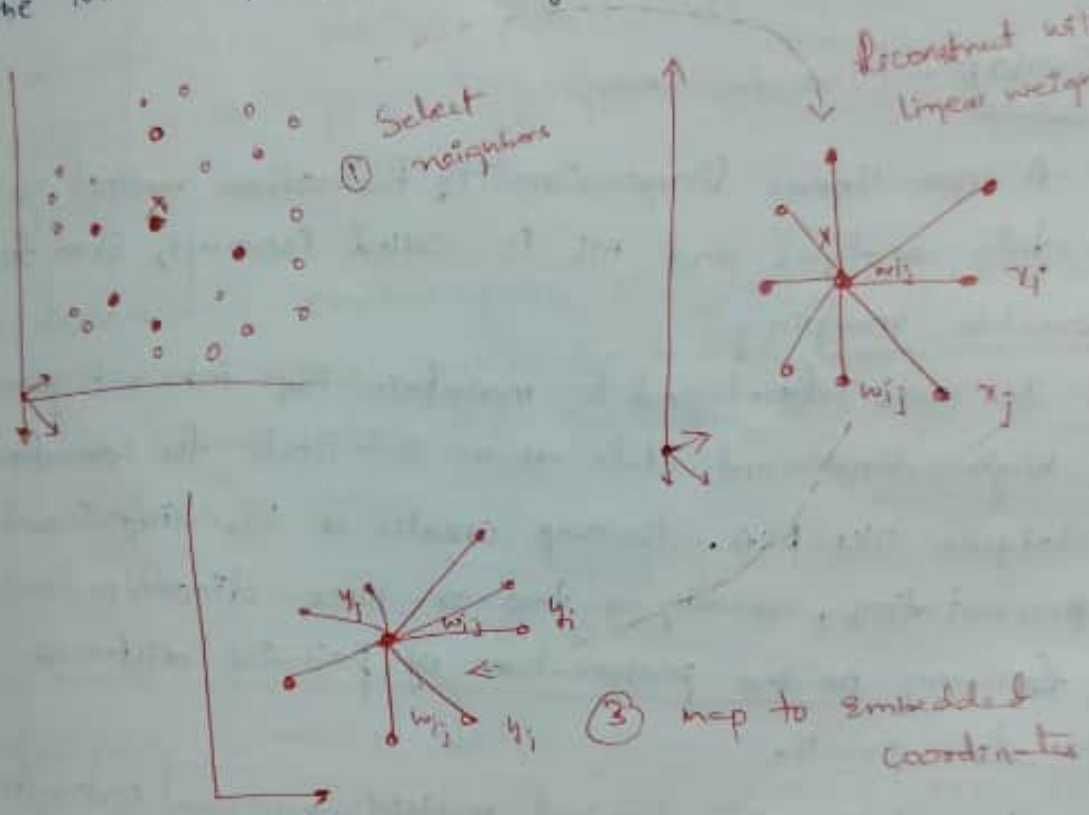
LLE was first proposed in [1,2].

Steps:

① Construct KNN Graph – Select neighbors

② Calculate the reconstruction of weights for reconstructing every point by its neighbors.

③ use the obtained weights to Embed the points in the low dimensional subspace.



Parameters

① K(No of neighbors) – It determines how many nearest neighbors are considered when Constructing the weight matrix. Selecting an appropriate value for K is Essential for the alg succes.

② Dimensionality Of Output Space –

You can specify the dimensionality of the lower dimensional space to which the data will be mapped:

③ Distance Metric – To define the proximity between Points. [Euclidian, Manhattan etc].

④ Optimization Algorithm –

It often uses optimization techniques like eigen vector methods to find the lower dimensional representation.

Isomap:– [Isometric mapping]

A non-linear Dimensionality Reduction method used in data analysis and ML is called isomap, short for isometric mapping.

It was developed to maintain the inherent geometry of high-dimensional data as a substitute for conventional techniques like PCA. Isomap creates a low-dimensional representation, usually a two-or-three-dimensional map, by focusing on the preservation of pairwise distances b/w data points.

It extends the classical multidimensional scaling (MDS) to nonlinear manifolds by incorporating geodesic distances.

It is particularly useful for uncovering the low-dimensional structure hidden in high-dimensional data.

It aims to preserve pairwise distance b/w points, is central to Isomap. It seek to achieve low dimensionality representation for the data while at the same time keeping geodesic distances.

Step-by-step procedure:

① Constructing the Neighborhood Graph:

- Identifying the k-nearest neighbors for each point in the dataset.

- connect each point to its KNN to form a graph.

② Computing shortest paths:

. Calculate the shortest path distances b/w all pairs of points in the graph. This involves finding the geodesic distances.

↓

To describe the shortest path b/w two points on a surface or within a space. unlike Euclidian distance, this distance accounts for the curvature and topology of the surface or space.

It is the length of shortest path connecting two points, considering the curvature.

③ Applying classical MDS:

- use classical Multidimensional Scaling (MDS) on the geodesic distance matrix to embed the data into a lower-dimensional space.

Eg: Consider a simple Dataset with 5 points in 2D space:

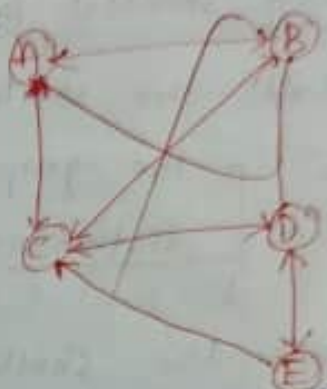
A(0,0), B(1,1), C(2,2), D(3,3), E(4,4)

Step1: Constructing the Neighborhood Graph

let choose k=2:

- Point A - NN are B & C
  Point B - NN are A & C
  Point C - NN are B & D
  Point D - NN are C & E
  Point E - NN are D & C

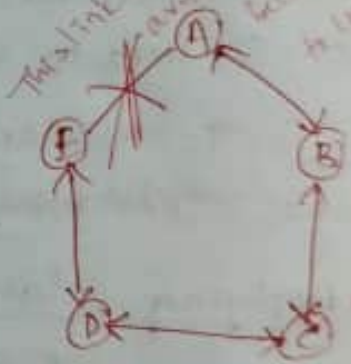
The neighborhood graph is:



- A → B, C
  B → A, C
  C → B, D
  D → C, E
  E → D, C

Step2: Computing shortest paths distances b/w all pairs of points

	A	B	C	D	E
A	0	1	2	3	4
B	1	0	1	2	3
C	2	1	0	1	2
D	3	2	1	0	1
E	4	3	2	1	0



Step3: Applying classical MDS

Perform classical MDS on the distance matrix. Here's a simplified version:

(1) Compute the centering matrix H and the Gram matrix B.

(2) Perform eigen decomposition of B to obtain the top eigenvalues and eigenvectors.

⑤ The coordinates in the lower-dimensional space are given by the top eigenvectors scaled by the square roots of their corresponding eigenvalues.

In this example, since the data points already lie on a straight line 2D, the lower-dimensional embedding will accurately reflect this linear structure.

Visualization:-

The original points and the transformed points are essentially the same in this simplified example, as the points are already in a low-dimensional space.

MDS: Multidimensional scaling (MDS) is a powerful technique used to analyze and visualize the similarity or dissimilarity of data points in a high-dimensional space by projecting them into a lower-dimensional space.

Least Squares Optimization:-

It is a fundamental technique in statistical regression analysis and machine learning, used to minimize the diff b/w observed data points and a models predictions.

It essentially seeks to find the best-fit line (or curve) to a set of data points by minimizing the sum of the squares of the residuals [the diff b/w observed & predicted values].

This method ensures that the overall Error is reduced, providing a highly accurate model for predicting future data trends.

In statistics, when the data can be represented on a Cartesian plane by using the independent & dependent variable as the x & y coordinates, it is called 'scatter data'.

This data might not be useful in making interpretations or predicting the values of the dependent variable for the independent variable. So, we try to get an eq^n of a line that fits best to the given data points with the help of the least Square Method.

It is used to derive a generalized linear eq^n b/w two variables. When the value of the dependent and independent variable is represented as the x and y coordinate in a 2D Cartesian coordinate system.

Initially, known values are marked on a plot. The plot obtained at this point is called a Scatterplot.

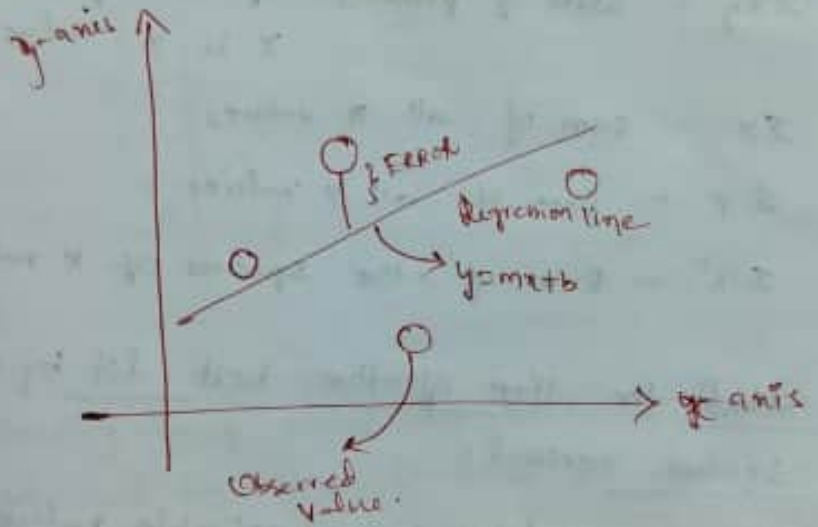
Try to represent all the marked points as a straight line or linear eq^n. The eq^n of such a line is obtained with the help of the least Square method.

This is done to get the value of the dependent variable for an independent var for which the value was initially unknown. This helps to make predictions to the value of dependent variable.

Definition:-

It is a statistical technique used to find the equation of best-fitting curve or line to a set of data points by minimizing the sum of the squared differences b/w the observed values and the values predicted by the model.

This method aims at minimizing the sum of the squares of deviations as much as possible. The line obtained from such a method is called a regression line or line of best fit.



Formula:-

This formula is used to find the best-fitting line through a set of data points.

For a simple linear regression, which is a line of the form $y=mx+c$, where

y - dependent variable
x - independent variable
a - slope of the line

b - y-intercept,

the formulas to calculate the slope (m) and the intercept (c) of the line are derived as

Slope (m): $m = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$

Intercept (c): $c = \frac{(\Sigma y) - a(\Sigma x)}{n}$

where n - no of data points

Σxy - Sum of product of each pair of x & y values.

Σx - sum of all x values

Σy - sum of all y values

Σx^2 - sum of the squares of x values.

Steps to find the line of the best fit by using the least square method:

Step1: Denote the independent variable values as 'x_i' & the dependent ones as 'y_i'. [formulate model]

Step 2: → Compute residuals
Calculate the avg values of x_i & y_i as X & Y.

Step3: presume the eq^n of the line of best fit as $y = mx + c$, where 'm' is the slope of the line 'c' represents the intercept of the line on the y-axis

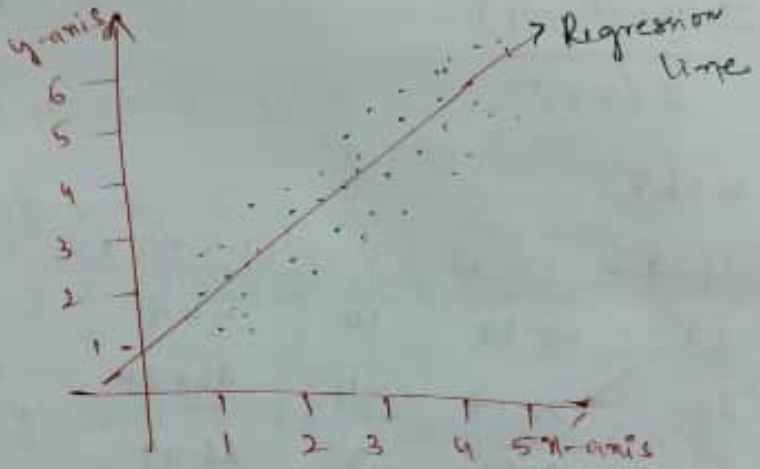
Step4: The slope 'm' can be calculated from the following formula. [minimize the sum of squared residuals]

$$m = \frac{\Sigma (x - x_i) \times (y - y_i)}{\Sigma (x - x_i)^2}$$

Step 5: The intercept 'c' is calculated from the following formula.

$$c = y - mx$$

Thus, we obtain the line of the best fit as $y = mx + c$,



Eg:

The following table shows the MidTerm and final Exam grades obtained by the students in database Course. Use the method of least squares to find the eq for the prediction of student's final exam grade based on the student's midterm grade in the course.

(x) Mid Term	(y) Final Exam
72	84
50	63
81	77
74	78
94	90
86	75
59	49
83	79
65	77
33	52
88	74
81	90

→ Predict the final exam grade of a student who received an 86 in the midterm exam

- Here, X - Independent variable
Y - dependent variable

- Eq^n of linear regression looks like: $y = a + bx$

where a - intercept
b - slope of coefficient

Q. How to cal value of a & b.

$$b = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\Sigma (x-\bar{x})^2}$$

$$\bar{y} = a + b\bar{x}$$

(x) Hrs taken	(Y) finishers	$(X-\bar{x})$	$(y-\bar{y})$	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$
72	84	-0.17	10	-1.7	0.03
58	63	-22.17	-11	243.87	491.51
81	77	8.83	3	26.49	77.97
74	78	1.83	4	7.32	3.35
94	90	21.83	16	349.28	476.55
86	75	13.83	1	13.83	191.27
59	49	-13.17	-25	329.25	173.45
83	79	10.83	5	54.15	117.29
65	71	-7.17	3	-21.51	51.41
33	52	-39.17	-22	861.74	1534.29
88	74	15.83	0	0	250.59
81	90	8.83	16	141.28	77.59
				2004	3445.67

$\bar{x} = 72.17$ $\bar{y} = 74$

$$b = \frac{2004}{3445.67}$$

$$\boxed{b = 0.5816}$$

$$\bar{y} = a + b\bar{x}$$

$$\Rightarrow a = \bar{y} - b\bar{x}$$

$$a = 74 - (0.5816)(72.17)$$

$$\boxed{a = 32.02}$$

Cal value of a & b

$$\bar{y} = a + b\bar{x}$$

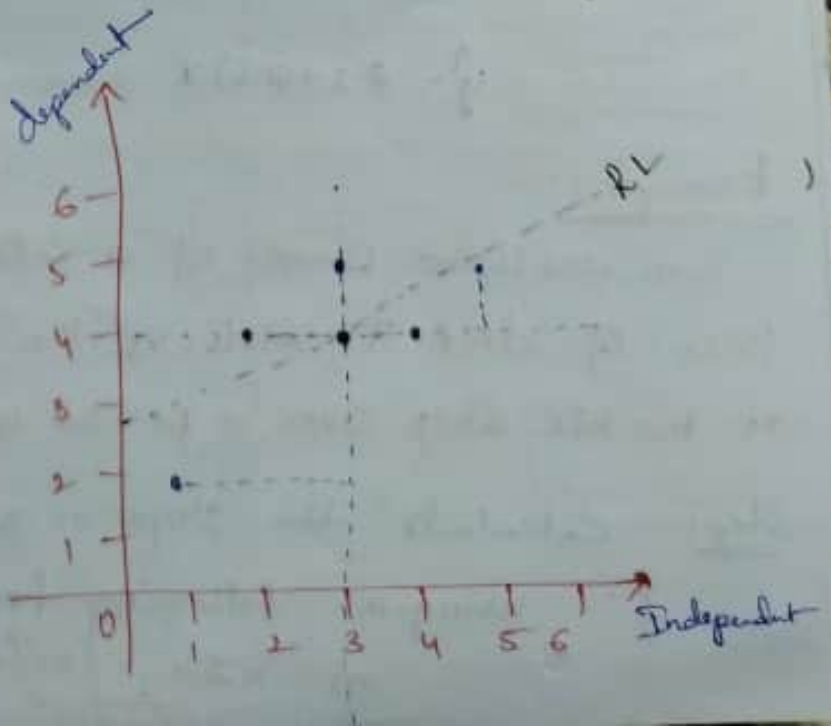
$\bar{y} = 32.02 + 0.5816\bar{x}$ → best fitting line

$$\Rightarrow y = 32.02 + 0.5816(86)$$

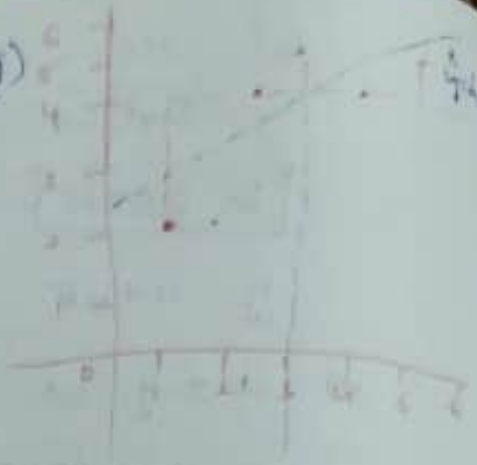
$\boxed{y = 82.04}$ → If the mid term grades are 86, the final term grades are 82.04

Second example:

X	Y
1	2
2	4
3	5
4	4
5	5
3	4



x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$	$(x-\bar{x})(y-\bar{y})$
1	2	-2	-2	4	4
2	4	-1	0	1	0
3	5	0	1	0	0
4	4	1	0	1	0
5	5	2	1	4	2
3	4			10	6



$$\bar{y} = \frac{6}{10} = 0.6$$

$$\text{Slope} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}$$

$\Rightarrow \hat{y} = b_0 + b_1 x$

$4 = b_0 + (0.6)(3)$

$b_0 = 4 - 1.8$

$\boxed{b_0 = 2.2}$

$\hat{y} = 2.2 + (0.6)x$

It is used to find the values of the coefficients that minimize the sum of the squared residuals.

Example 1:-

Tom who is the owner of a retail shop, found the price of diff. T-shirts vs the no of T-shirts sold at his shop over a period of one week.

Step1 Calculate the slope 'm' x axis by using the following formula

$$m = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

Price of T-shirts in dollars (x)	Sum of T-shirts sold (y)	$y = mx + c$	error
2	4	3.3	-0.67
3	5	4.9	-0.14
5	7	7.9	-0.89
7	10	10.9	0.93
9	15	13.9	-1.03
5.2	8.2		

$m = 1.518$ approximately

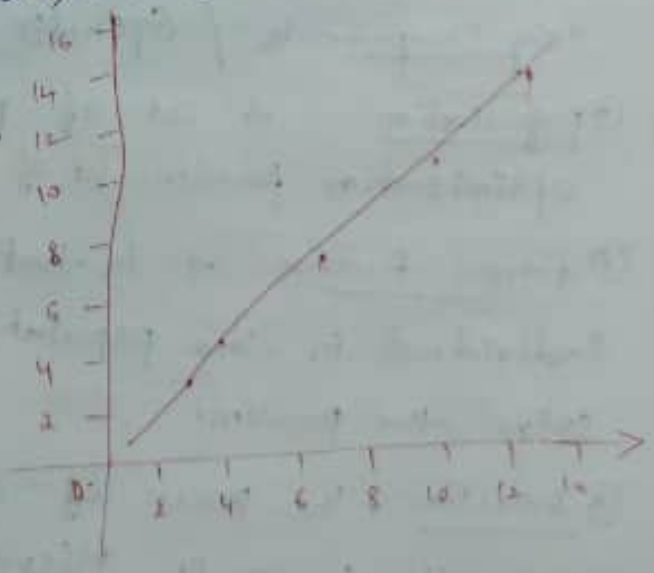
Step 2: Compute the y-intercept value $[c = y - mx]$

$$c = 8.2 - (1.518)(5.2)$$
$$= 8.2 - 7.89$$
$$c = 0.305$$

Step 3: Substitute the values in the final eqn.

$$y = mx + c$$

$$\Rightarrow (1) - y = mx + c = (1.518)(8.2) + (0.305)$$



How Tom can use the above eqn to estimate how many T-shirts of price $8 can be sell at the retail shop.

$$y = 1.518(8) + 0.305$$
$$= 12.45 \text{ (T-shirts)}$$

This comes down to 13 T-shirts

→ Evolutionary learning :-

It is a fascinating area of Machine learning that draws inspiration from the process of natural evolution. It involves using evolutionary algorithms to optimize and solve complex problems by mimicking biological mechanisms such as selection, mutation, and reproduction.

Evolutionary Algorithms are a subset of AI and optimization techniques that mimic the proces of natural evolution.

These algorithms are inspired by biological mechanism like selection, mutation, recombination(crossover) and survival of the fittest.

These are particularly useful for solving complex optimization problems where traditional methods might fail.

Key Components / Genetic Operators

① Population:- A set of potential solutions to the optimization problem [set of ind which called as population]

② fitness function: A fn that Evaluates and scores each Individual in the population based on how well they solve the problem.

③ Selection: The proces of choosing individuals from the population to create offspring for the next generation. Higher fitness individuals are more likely to be selected.

④ Crossover (recombination): Combining parts of two parent soln to create offspring.

⑤ Mutation: Introducing random changes to an individual to maintain genetic diversity

⑥ Replacement - Deciding which individuals from the current generation will be carried over to the next generation.

Genetic Algorithm:-

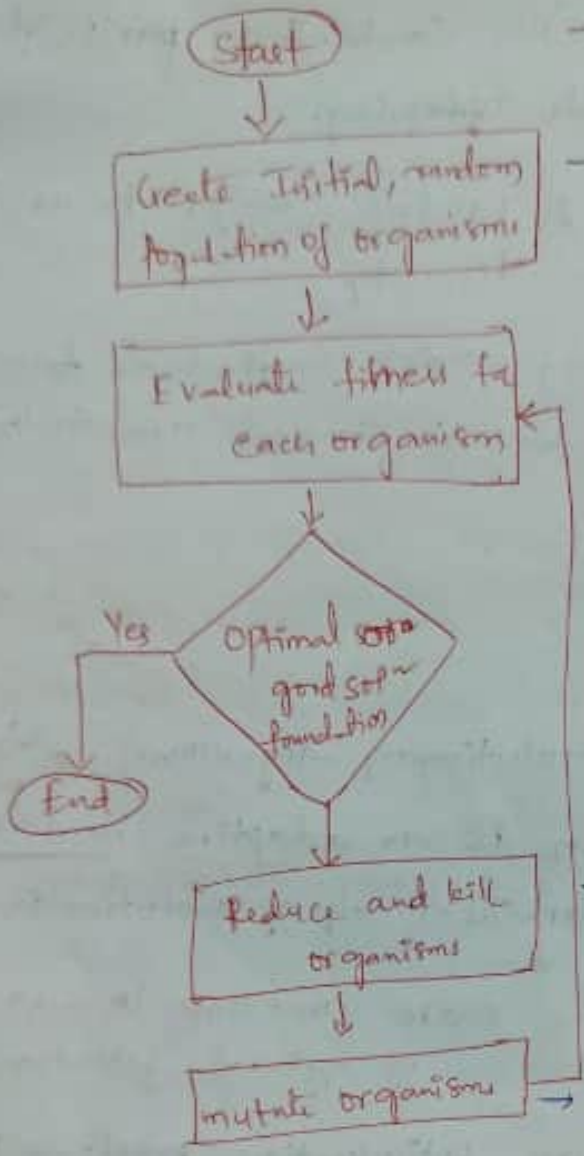
- They belong to Evolutionary Algorithms with respect to no of parameters & type of parameters
- A genetic algorithm is an adaptive heuristic search alg inspired by "Darwin's theory of evolution in nature". changing according to situation
  - How species evolve over time through a process of natural selection.

- It is used to solve optimization problems in ML.
- It helps to solve the complex problems that would take a long time to solve.
- It is based on genetics and natural selection.
- used to generate high quality soln for an optimization problem.

Applications:

1) Designing Electronic circuits
2) Code-breaking
3) Image processing
4) Artificial creativity.

-How genetic alg works



→ we have to create (list) randomly create some population

→ whichever org having small values. kill them and replace with higher values

→ making changes & evaluate.

Operations:-

① Selection - selected among all the samples
from c → select 2 most fittest parents

② Crossover - Reproduction — new offspring

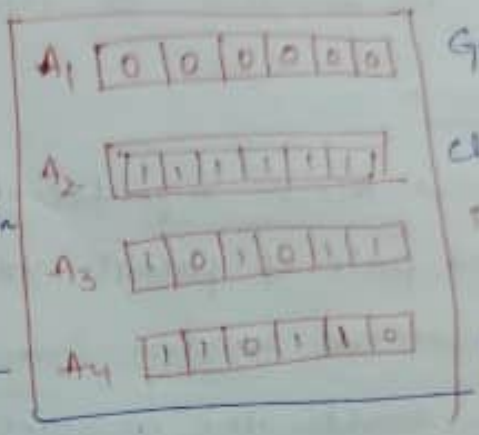
③ Mutation — make some changes | probability of getting more accurate results | optimal solutions

④ Encoding -

- This alg reflects the proces of natural selection where the fittest individuals are selected for reproduction in order to produce offspring of the next generation

① Initial population:

- proces begins with a set of individuals which is populn
- Each individual is a sol^n to the problem you want to solve known as Chromosome



- An ind is characterized by a set of parameters (var) known as Genes.
- Genes are joined into a string to form a chromosome (solution)

② Fitnes function:

- It determines how fit an individual is? (Ability of an individual to compete with other individuals)
- It give a fitnes score to each individual.
- The probability that an individual will be selected for reproduction is based on fitnes score.

③ Selection:

- Idea of a selection phase is to select the fittest individuals and let them pass their genes to the next generation.

- Two pairs of individuals (parents) are selected based on their fitness scores
- Individuals with high fitness have more chance to be selected for reproduction.

④ Crossover -

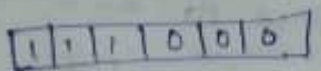
- It is the most significant phase in a genetic alg.
- for each pair of parents to be mated, a crossover point is chosen at random from within the genes
- For eg, consider the crossover point to be 3.

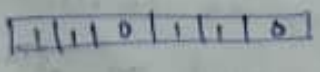
⑤ Offspring

- are created by exchanging the genes of parents among themselves until the crossover point is reached.
- The new offspring are added to the population.

⑥ Mutation:

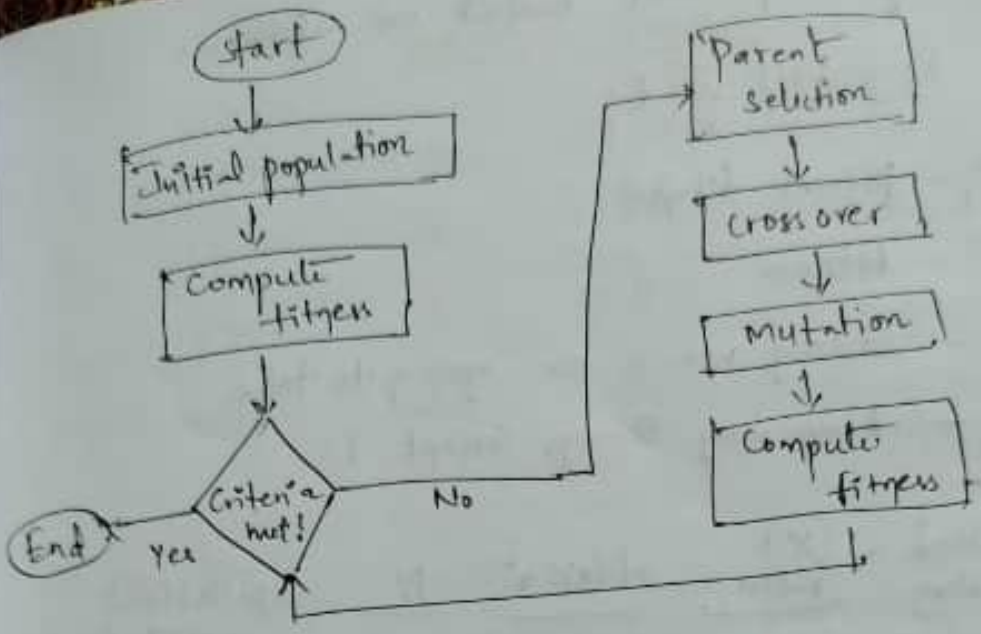
- new offspring formed, some of their genes can be subjected to a mutation with a low random probability.

- Before - 

After - 

⑦ Termination:

- The alg terminates if the popul has converged (does not produce offspring which are significantly different from the previous generations).



Example:- [Roulette wheel selection]

$f(x) = x^2$, maximise this function with x in interval [0, 31].

① Generate initial population at random (Called genotypes)

N = 4 → organisms

13 - 01101
24 - 11000
8 - 01000
19 - 10011

} binary representation

② Calculate fitness - (a) Decode into an integer (phenotypes)

13 - 01101 24 - 11000
8 - 01000 19 - 10011

(b) $f(x) = x^2$

13 → 169, 24 → 576, 8 → 64, 19 → 361

Now, we have to select any 2 parents based on the fitness, we have to calculate the fitness of the parents.

(3) Select any 2 parents based on fitness.

$$P_i = F_i / \sum_{j=1}^{n} F_j$$

P_i - parent fitness

F - fitness

i - string which are going to take

j - Remaining strings except 'i'.

No	Initial Population	(X) Value	$f(x) = x^2$	P_i	expected Count = NXP [P_i xn]
1	01101	13	169	0.14	0.56
2	11000	24	576	0.49	1.97 → highest
3	01000	8	64	0.06	0.22 → Smallest
4	10011	19	361	0.31	1.23
			1170	1.00	4.00

① $P_i = \frac{13}{1170} = 0.14$

(4) For Crossover, - we need to select parents

↓

Can be either one point / 2 point / n point

eg 100|1101 ⇒ 10001001

101|01001 ⇒ 10111101

for 2 point - 100|111|01 ⇒ 100|010|01

101|010|01 ⇒ 10111101

Here, String 2 [highest] 3 [lowest] - Instead of writing 3 values we are eliminating 3 & replace with it '2'.

NO	Initial	Crossover Point	After Crossover	x	$f(x) = x^2$
1	0110|1	4	01100	12	144
2	1100|0	4	11001	25	625
3	11|000	2	11011	27	729
4	10|011	2	10000	16	256
					1754

(5) Mutation

- Applied to each & every child individually after Crossover.
- Bits are changed from '0' to '1' or from '1' to '0' at randomly chosen position of randomly selected strings.

NO	After Crossover	After Mutation	x	$f(x) = x^2$
1	01100	11100	28	784
2	11001	11001	25	625
3	11011	11011	27	729
4	10000	10100	20	400
				2538

→ Each and Every the value is going to be increased.

- It does not need any derivative info.